

- 25** The DNA triplets of genes are translated as amino acids or stop signals during protein synthesis. The table shows some of these triplets.

DNA triplet	name of amino acid
ATA	tyrosine
ATG	tyrosine
ACA	cysteine
ACG	cysteine
ATC	<i>stop signal</i>
ACC	tryptophan

What could be the effects of **one** substitution mutation in a triplet coding for tyrosine?

- 1 The triplet is translated as cysteine.
- 2 The triplet is translated as tryptophan.
- 3 The triplet is translated as tyrosine.
- 4 Translation stops at this triplet.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4